Business Intelligence – Lab 1

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Abstract:

Nota Bene: This document is designed as an autonomous capsule. It contains lecture parts – explaining concepts – and lab parts – practical applications through coding exercises.

⚠️ **Important remark**: This capsule will use a set of icons and graphical chart to indicate several levels of information.

- 👉: Something that you should **do** to complete the Lab
- 💾: You need to **validate** something (call the teacher, save something for deposit)
- 💡: Tips or **useful** remark to help you in your lab
- 🧠: An explanation on a new knowledge, that you should probably know/**retain**
- ⚠️: Something important that you **need** to read
- ⭐: Something to try if you want to **explore** further on the subject
- IT SHOULD NOT BE NEEDED TO BE WRITTEN BUT EVERYTHING ON WHITE BACKGROUND SHOULD BE READ ALSO! (Only ⭐ is optional).

Preamble: Project environment

Welcome to your **Business Intelligence capstone project**!
Throughout this module, you won't just *learn* BI — you will **build your own BI solution**, just like in a real Data team.
You will work **in pairs** and progress step by step during the lab sessions.

🎯 The mini-project concept

During the project, you will define a business need (your choice), select a dataset, and then build:

- A requirements document
- An agile backlog
- A complete BI pipeline (ingestion → transformation → model)
- A final dashboard

By the end, you will be able to lead a BI project from A to Z, just like in a company.

💡 Lab 1 – Dataset Selection & Requirements Document

Task 1: Choose a dataset

👉: Find a relevant dataset for your project

Constraints:

1) Minimum 500 rows and a final semantic model containing at least 3 tables.

(You can either choose a multi-table dataset or a big single table that will be split into a fact table and dimensions).

2) Format: CSV, Excel…

3) Free domain: sports, finance, environment, transport, video games, movies, health, etc.

Data should be simple but usable for:

- at least **2–3 KPIs**

- at least **1–2 impactful visualisations**

⚠️ **Important**
Take your time to choose a dataset that will stay relevant throughout all the TPs.
You will create visualisations with it later**, so pick something interesting.**

💡 : You can find dataset on:

- Kaggle
👉 https://www.kaggle.com/datasets

- Microsoft Learn – Sample Datasets (AdventureWorks, Contoso)

👉 https://learn.microsoft.com/en-us/power-bi/sample-datasets

👉 Create a Word file named NAME1_NAME2_TP1_BI.docx

👉 Add to your file the context of your project : Write a project context explaining the situation in which your dashboard will be used.

Your text should answer the following questions:

- What business problem is your client facing?

- Why do they need a BI solution?

- What data is available, and why is it insufficient in its current state?

- What type of decisions should this dashboard help to make?

🧠 **Objective:** create a clear narrative framework that will justify your future KPIs, User Stories, and backlog.

👉 Add to your file the Target User: Define who the end user of your dashboard will be.

Specify:

- their role (manager, analyst, executive, etc.),

- their needs,

- their level of data expertise,

- and why they need an overview.

👉: Add to your file:

- The link to the dataset source

- A summary table including:

    Table names

    Number of rows

    Column dictionary (max 3–4 lines)

    Perceived data quality (duplicates? NaN values? granularity?).

⚠️ **Reminder**
You will deposit your file on Moodle at the end of the lab.
Make it clean and readable. Explain your reasoning and the steps of your work.

⚠️ : **Don’t forget to cite any AI generated code**.

Task 2: Define the business need

👉 Create a client and a business scenario, add it to your Word file

💡 **Examples:**

- A company wants to track its sales
- A city wants to monitor air quality
- A sports federation wants to track its athletes' performance
- An HR team wants to analyze recruitment

Mandatory questions to address:

- What is the problem to solve?
- Who is the end user? (director, analyst, manager, etc.)
- Which decisions should be supported?
- Which KPIs will be needed?

💬 **Objective: make the business need concrete — not just *"we're going to build a dashboard."***

Task 3: Write the requirements/specifications document

👉 Add to your file the requirements/specifications of your project

Required structure:

1. Context & objective

Around ten lines explaining the business need, the users, and the objective.

2. Scope (in scope / out of scope)

• What you will do

• What you will not do

3. Planned KPIs + definitions

3 to 5 KPIs maximum, with a formula or clear logic.

4. Data used

• Tables

• Granularity

• Possible limitations

5. Identified constraints

• Data quality

• Volume

• Missing values

• Units / dates / data types

⭐ **Final deliverable of Lab 1**

At the end of the lab, your Word file must contain:

1. Dataset summary
2. Business need scenario
3. Requirements document

💾 : Remember to upload your file on Moodle